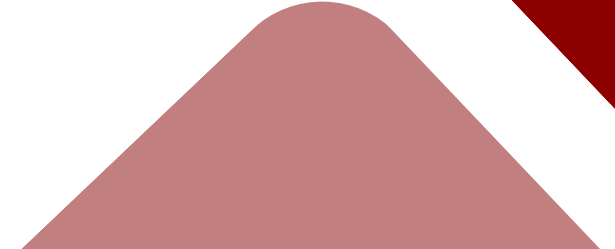




EXAMINATION **SYSTEM**



Our TEAM

Examination System



RAAFAT ELRAIS



OMAR MOHAMMED



YOUSIF SABAL



YOUSIF ASHRAF



AHMED MEDHAT

Powered by ITI

Table of Contents

- Problem Statement
- Objectives
- ER-Diagram & Schema
- Database implementation
- Stored Procedures
- Streamlit website
- SSRS Reports
- Power BI Dashboards
- Power Apps Exam Screen
- Problems and Solutions
- Used Tools
- Future Plan

What is **THE PROBLEM?**

Student exam processes are **traditionally**:

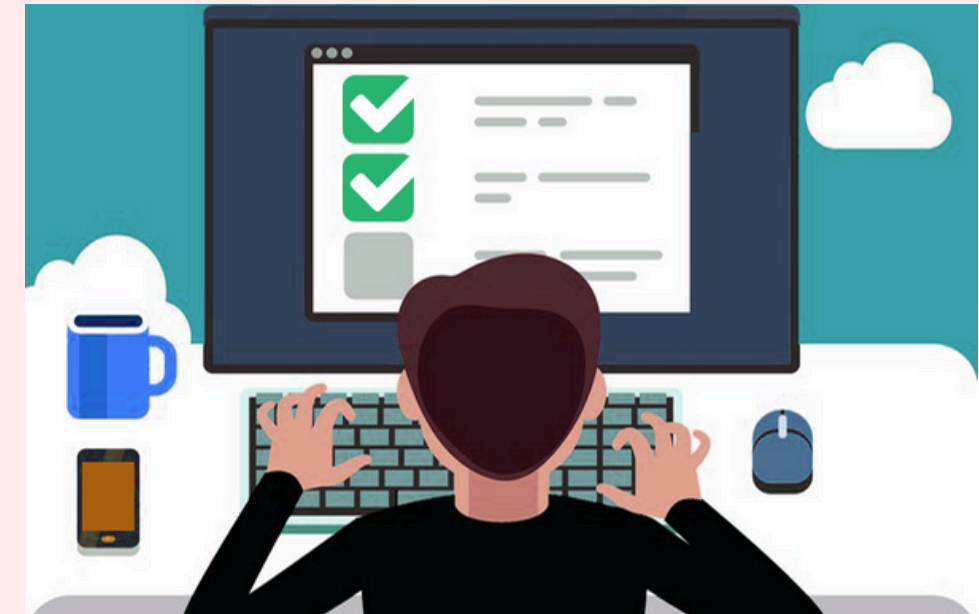
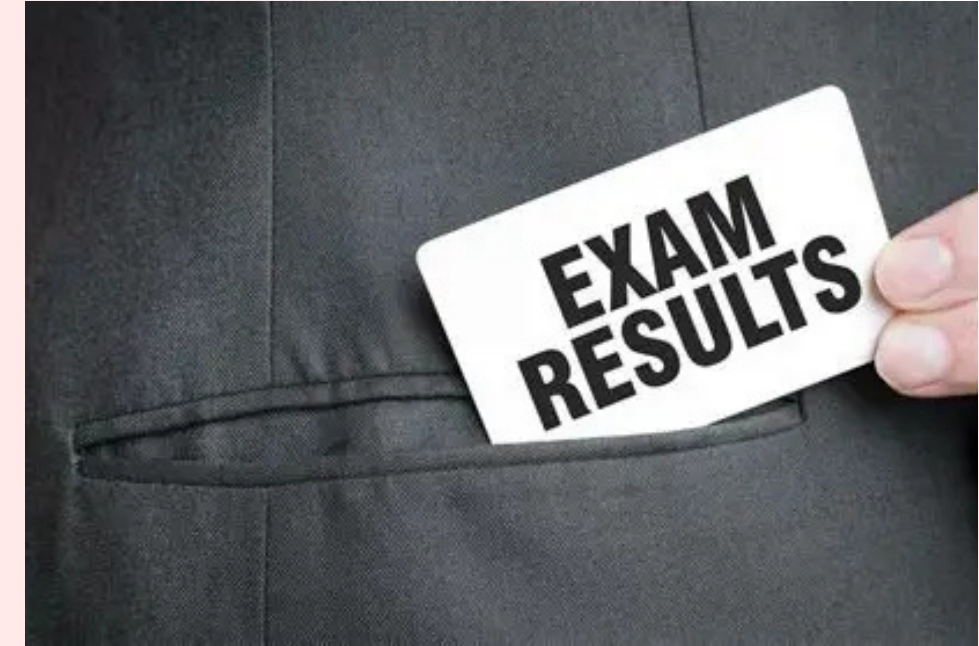
- No centralized tracking
- Difficult performance measurement
- No real-time insights for instructors
- No automatic scoring
- Difficult to analyze topic weaknesses

Academic institutions **Suffer** from:

- Human error in grading
- Delayed results
- Limited analytics

OBJECTIVE

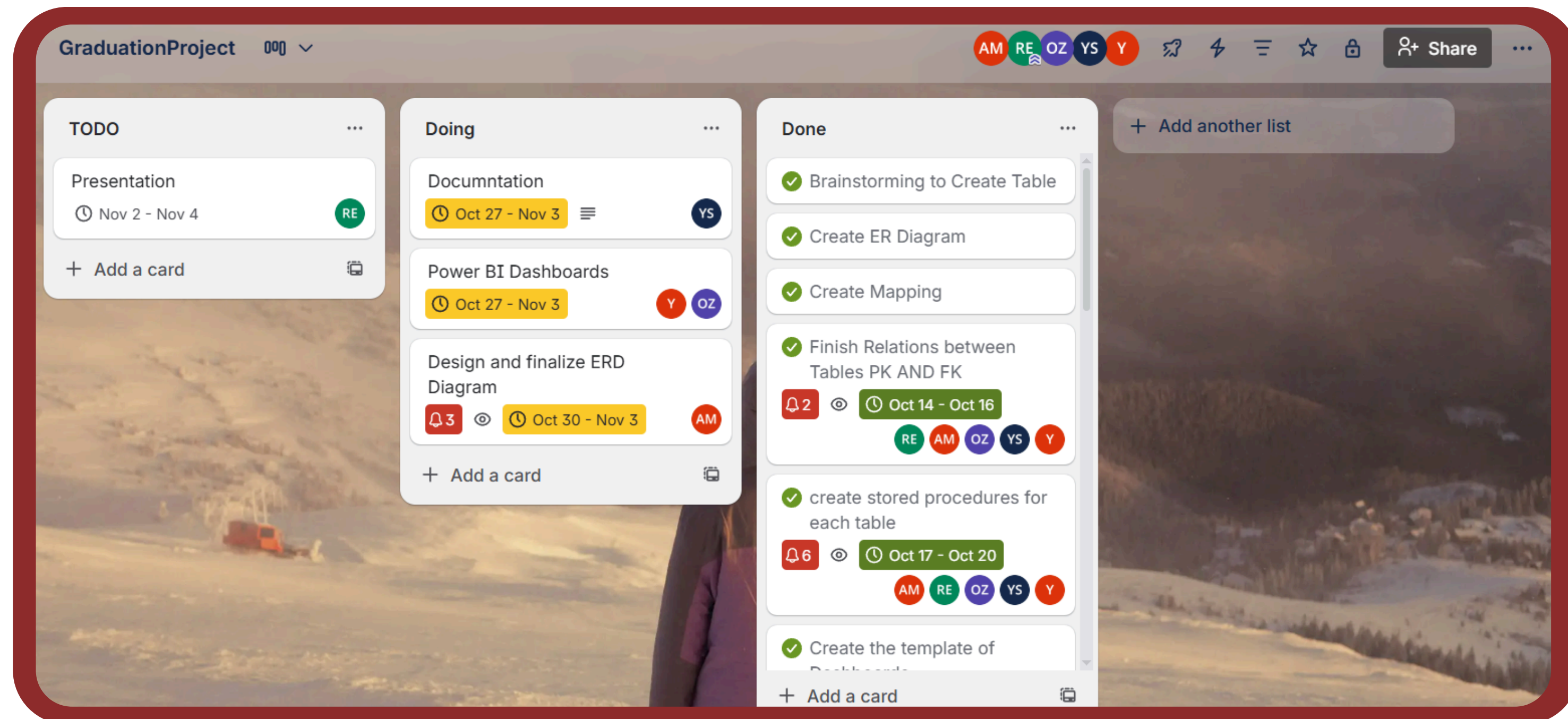
- Digitize the entire student examination process
- Automate grading
- Store answers securely
- Track performance over time



HOW DO WE MANAGE WORK AMONG US

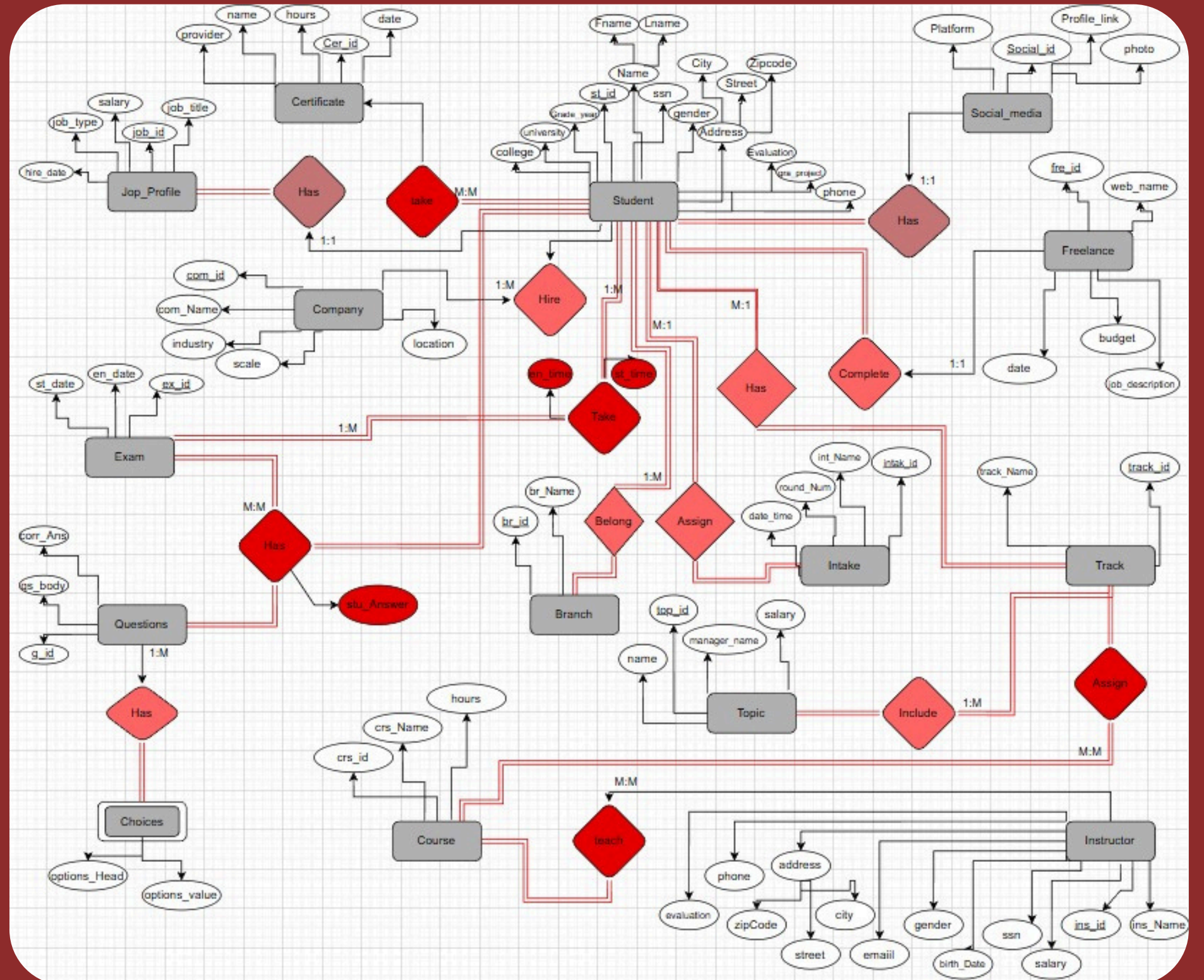
Methodology:

- Agile (2 Days - Sprint)
- Trello / Jira board



ER-DIAGRAM

- **We designed a relational database with carefully structured tables to ensure data integrity.**
- **Relationships follow a 1-to-Many model in most areas, such as students to exam answers, and courses to questions.**
- **Normalization minimized redundancy, while lookup tables improved consistency.**



MAPPING

- 15 Main tables
- 7 M:M TABLES

- **Student** (ST_ID **P.K**, SSN, Fname, Lname, Gender, Birth_Date, Age, Address (City, Street, Zip_Code), Phone, University, College, Grade_Year, Grade_project, Evaluation, Email, Password, com_Id **F.K**, pro_Id **F.K**, track_Id **F.K**, intake_Id **F.K**, Branch_Id **F.K**) MAIN
- **Company** (Com_ID **P.K**, Com_name, Location, Industry, Scale)
- **Project** (Pro_ID **P.K**, Pro_Name track_Id **F.K**)
- **Freelance** (Fre_ID **P.K**, Website_name, Date, Budget, Job_description, ST_ID **F.K**,)
- **Course** (Crs_ID **P.K**, Crs_Name, Hours)
- **Student_Certificate** (st_Id **F.K**, cer_Id **F.K**) **P.K**
- **Student_Exam** ((ex_Id **F.K**, st_Id **F.K**) **P.K**, start_Time, end_Time, score, Grade, Percentage)
- **Stu_Exam_Ques** ((EX_Id **F.K**, Qs_ID **F.K**, St_ID **F.K**) **P.K**, Stu_Answer, Question_Order)
- **Exam** (EX_Id **P.K**, Start_Time, End_Time, Crs_ID **F.K**)
- **Questions** (Qs_ID **P.K**, Qs_Body, Correct_answer, Q_Type, Crs_ID **F.K**)
- **Choices** (Qs_Id **F.K**, Options_Header, Option_Value)
- **Ins_Course** (crs_Id **F.K**, Ins_Id **F.K**) **P.K**
- **Track_Course** (Track_Id **F.K**, Crs_Id **F.K**) **P.K**
- **Branch_Track** (Track_Id **F.K**, Branch_Id **F.K**) **P.K**
- **Track** (Track_Id **P.K**, Name, Ins_Id **F.K**, topic_Id **F.K**)
- **Intake** (Intake_Id **P.K**, Start_Date, End_Date, Int_name, Round_num)
- **Certificate** (Cer_Id **P.K**, Name, Hours, Provider, date)
- **Social_Media** (Social_Id **P.K**, Platform, Profile_link, Photo, st_Id **F.K**)
- **Branch** (Branch_Id **P.K**, Br_Name)
- **Instructor** (Ins_ID **P.K**, Ins_Name, Salary, SSN, Birth_Date, Gender, Email, City, Street, ZIP_Code, PHONE, Evaluation)
- **Topic** (Topic_ID **P.K**, Name, Manager_Name, Salary)

DATABASE IMPLEMENTATION

1



Technology
Microsoft SQL Server

2



Data Types
Describe the data types used
for different fields (e.g., INT,
VARCHAR, DATE).

3



Schema Diagram
Outline the schema design,
including tables, relationships,
and constraints.

4



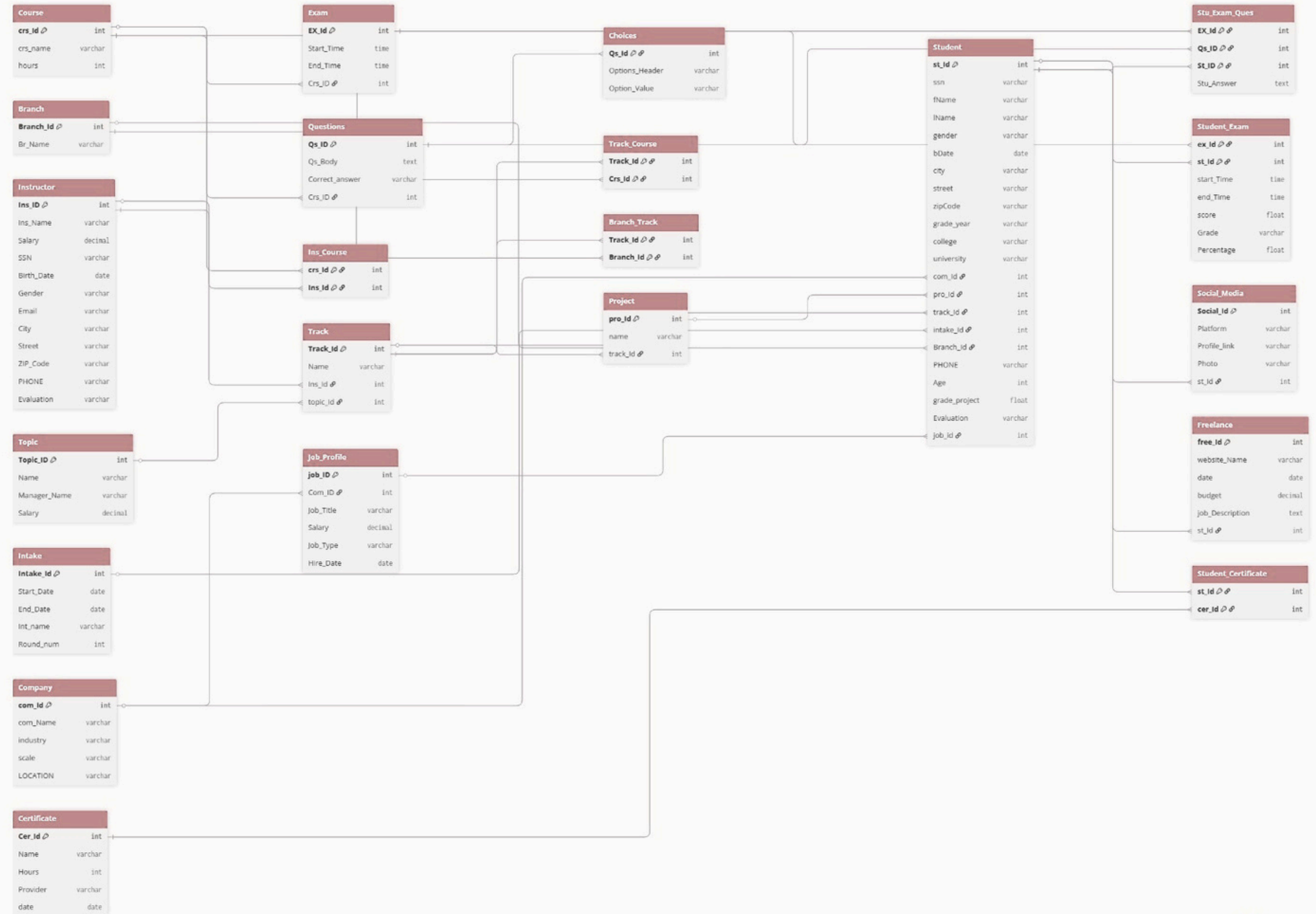
Security
Encrypted Database Objects
E.g: SP, Views

5



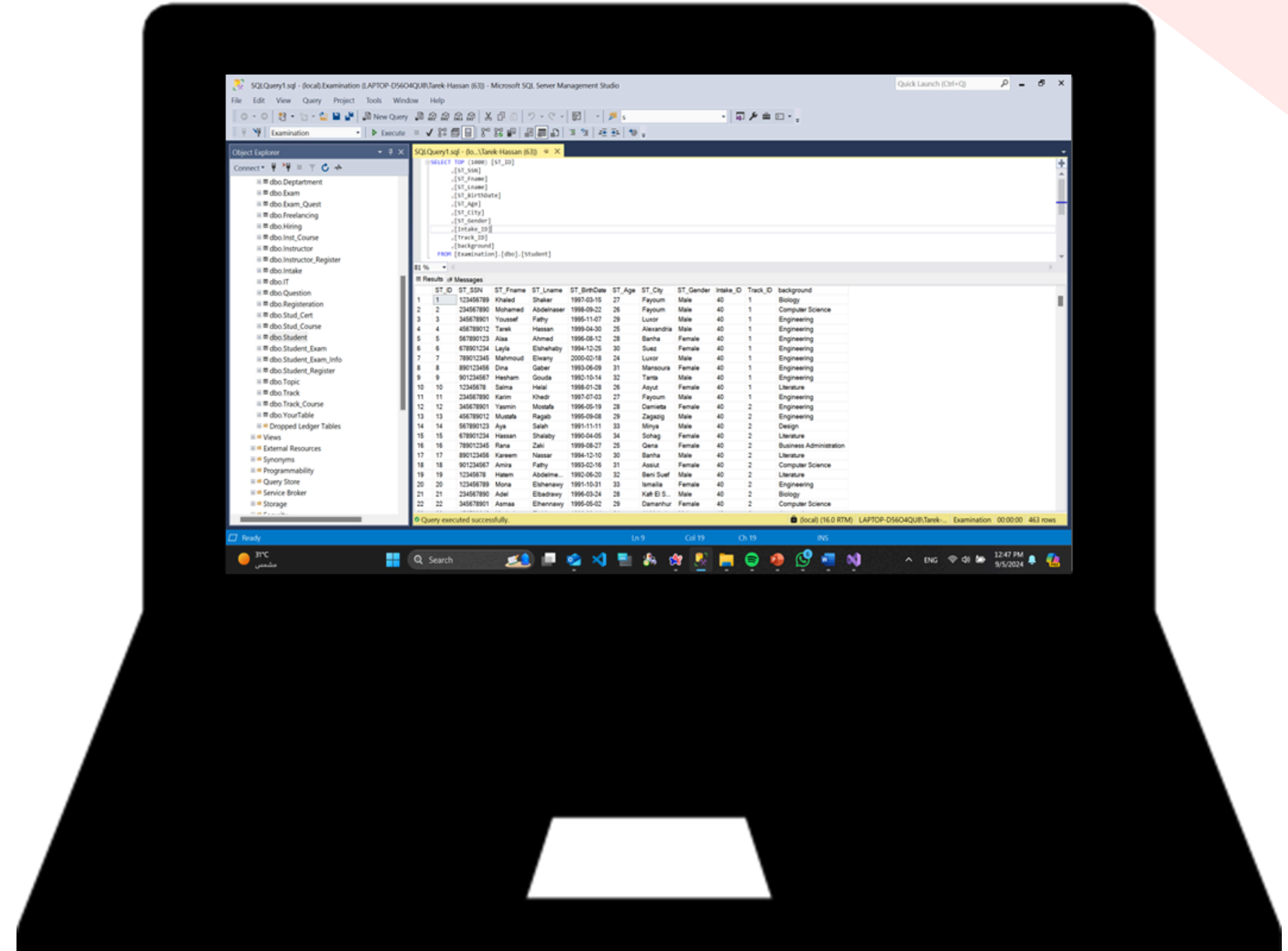
Backup & Recovery
Daily Backup

DATABASE DIAGRAM



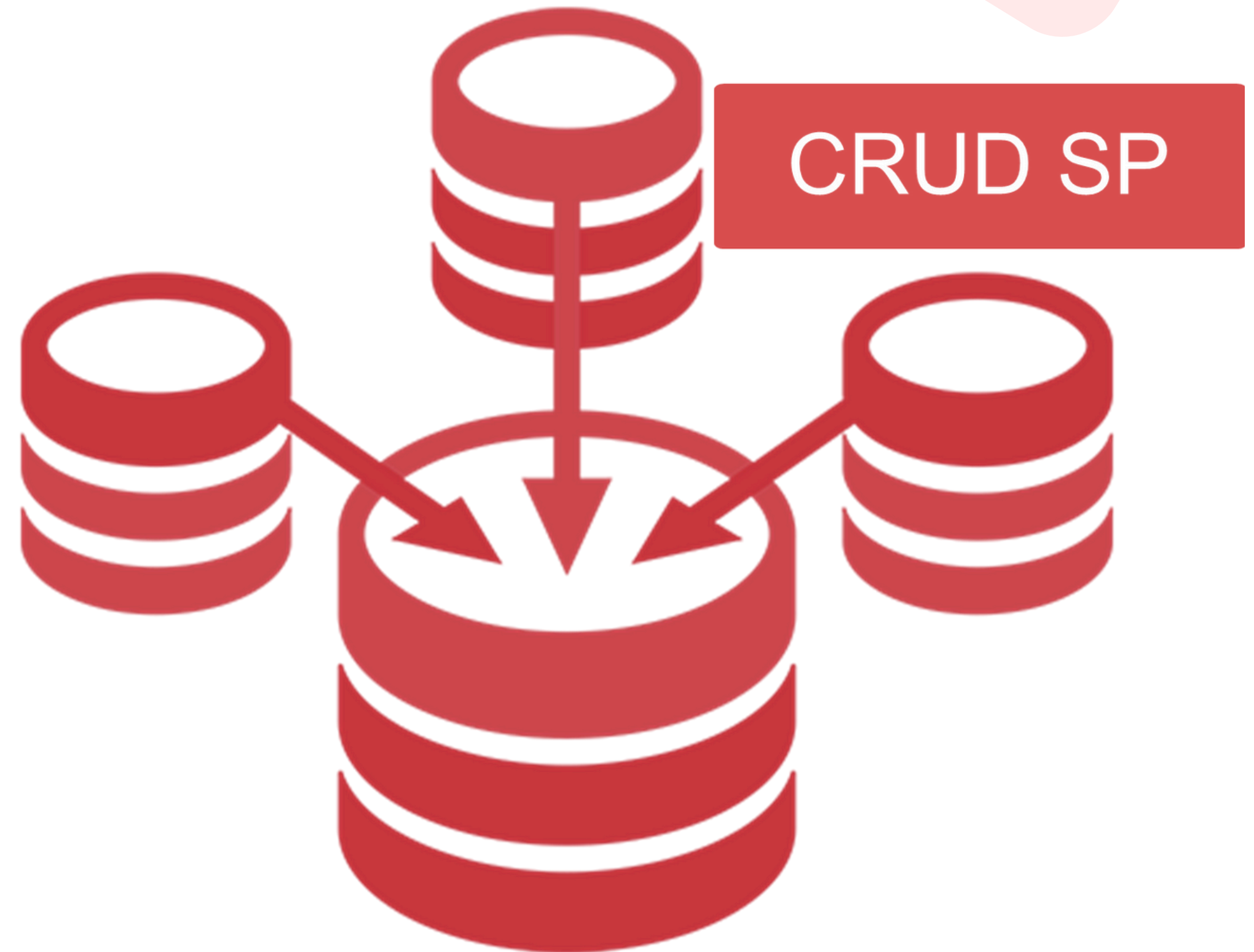
DATA FILL

- **AI-Generated**
(Claudera- Gemini – Mockaro – GPT)



CRUD STORED PROCEDURES

They simplify and secure database operations by encapsulating the logic for Create, Read, Update, and Delete (CRUD) actions for Each Table.



CRUD STORED PROCEDURES

Examples

SP For Selection

```
ALTER procedure [dbo].[SELECT_br_tr]
@BRANCH_ID INT= NULL,
@TRACK_ID INT = NULL
AS
BEGIN
SET NOCOUNT ON
SELECT
BRANCH_ID,
TRACK_ID
FROM Branch_Track
WHERE
(@Branch_ID IS NULL OR Branch_ID = @Branch_ID)
AND (@Track_ID IS NULL OR Track_ID = @Track_ID)
END
```

SP For Insertion

```
CREATE PROCEDURE [dbo].[INSERT_STUDENT]
@Fname NVARCHAR(100),
@Lname NVARCHAR(100),
@Gender NVARCHAR(10),
@Birth_Date DATE,
@City NVARCHAR(50),
@Street NVARCHAR(100),
@ZipCode NVARCHAR(10),
@Phone VARCHAR(11),
@Grade_Year VARCHAR(10),
@Com_Id INT,
@Pro_ID INT,
@Track_ID INT,
@Intake_ID INT,
@Social_ID INT,
@Branch_ID INT,
@College VARCHAR(100),
@University VARCHAR(100),
@Grade_Project VARCHAR(30),
@Evaluation NVARCHAR(20)
AS
BEGIN
if Exists(select 1 from Student
where SSN=@SSN)
begin
print('This Student already exists');
return;
end;
begin try
IF NOT EXISTS (SELECT 1 FROM Student WHERE SSN = @SSN)
BEGIN
INSERT INTO Student
([SSN],[Fname],[Lname],[Gender],[Birth_Date],[City],[Street],
[ZipCode],[Phone],[Grade_Year],[Com_ID],[Pro_ID],[Track_ID],
[Intake_ID],[Social_ID],[Branch_ID],
[College],[University],[Grade_Project],[Evaluation])
VALUES
(@SSN,@Fname,@Lname,@Gender,@Birth_Date,@City,@Street,
@ZipCode,@Phone,@Grade_Year,@Com_Id,@Pro_ID,@Track_ID,
```


CRUD STORED PROCEDURES

SP For Deletion

```
ALTER PROCEDURE [dbo].[Delete_Student]
    @St_Id INT
AS
BEGIN
    BEGIN TRY
        DELETE FROM Student
        WHERE St_Id = @St_Id;

        PRINT ('Student deleted successfully')
    END TRY
    BEGIN CATCH
        PRINT ('Cannot delete Student')
    END CATCH
END;
```

SP For Update

```
    @Pro_ID INT,
    @Track_ID INT,
    @Intake_ID INT,
    @Social_ID INT,
    @Branch_ID INT,
    @College NVARCHAR(100),
    @University NVARCHAR(100),
    @Grade_Project VARCHAR(30),
    @Evaluation NVARCHAR(20)
AS
BEGIN
    UPDATE Student
    SET
        [Fname] = @Fname,
        [Lname] = @Lname,
        [Gender] = @Gender,
        [Birth_Date] = @Birth_Date,
        [City] = @City.
```

EXAM STORED PROCEDURES

1 Add New Exam

```
ALTER PROCEDURE [dbo].[Add_Exam]
    @Start_Time TIME,
    @End_Time TIME,
    @Crs_ID INT
AS
BEGIN
    SET NOCOUNT ON;

    DECLARE @Today DATETIME = CAST(GETDATE() AS DATE);

    INSERT INTO Exam ( Start_Time, End_Time, Crs_ID, Exam_Date)
    VALUES ( @Start_Time, @End_Time, @Crs_ID, @Today);

    DECLARE @New_Ex_ID INT = SCOPE_IDENTITY();
    PRINT 'Exam Added Successfully. Exam ID: ' + CAST(@New_Ex_ID AS NVARCHAR(10));
END;
```

2 Register New Student

```
ALTER proc [dbo].[Register_Student] @St_ID INT , @Ex_ID INT
AS
BEGIN
    Set Nocount on;
    If Not Exists (Select 1 from Student_Exam where St_ID = @St_ID and Ex_ID = @Ex_ID)
    begin
        insert into Student_Exam (St_ID , Ex_ID , Start_Time , End_Time)
        select @St_ID , @Ex_ID , Start_time , End_time
        From Exam
        where Ex_ID = @Ex_ID
        Print 'Student Registered For Exam'
    end
    else
    begin
        Print 'Student Already Registered'
    end
end;
```

EXAM STORED PROCEDURES

3 Generate Random Questions

```

@St_ID INT,
@Crs_ID INT
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT EXISTS (SELECT 1 FROM Student_Exam WHERE St_ID = @St_ID AND Ex_ID = @Ex_ID)
    BEGIN
        PRINT 'Student not registered for this exam.';
        RETURN;
    END
    CREATE TABLE #temp_questions
    (
        Temp_ID INT IDENTITY(1,1) PRIMARY KEY,
        Qs_ID INT,
        Qs_Body VARCHAR(1000)
    );
    INSERT INTO #temp_questions (Qs_ID, Qs_Body)
    SELECT TOP 10 Qs_ID, Qs_Body
    FROM Questions
    WHERE Crs_ID = @Crs_ID
    ORDER BY NEWID();

    INSERT INTO Stu_Exam_Qs (Stu_ID, Ex_ID, Qs_ID, Question_Order)
    SELECT @St_ID, @Ex_ID, Qs_ID, Temp_ID
    FROM #temp_questions;
    SELECT Temp_ID AS QuestionNumber, Qs_Body AS Question
    FROM #temp_questions
    ORDER BY Temp_ID;

```

4 Submit Answers and Calculate Scores

```

ALTER PROCEDURE [dbo].[SubmitExamAnswers]
    @St_ID INT,
    @Ex_ID INT,
    @Answers NVARCHAR(MAX)
AS
BEGIN
    SET NOCOUNT ON;
    ;WITH StudentAnswers AS (
        SELECT
            value AS Stu_Answer,
            ROW_NUMBER() OVER (ORDER BY (SELECT NULL)) AS rn
        FROM STRING_SPLIT(@Answers, ',')
    )
    UPDATE S
    SET S.Stu_Answers = SA.Stu_Answer
    FROM Stu_Exam_Qs S
    INNER JOIN StudentAnswers SA
        ON SA.rn = S.Question_Order
    WHERE S.Stu_ID = @St_ID AND S.Ex_ID = @Ex_ID;
    DECLARE @Correct INT;
    SELECT @Correct = COUNT(*)
    FROM Stu_Exam_Qs S
    JOIN Questions Q ON S.Qs_ID = Q.Qs_ID
    WHERE S.Stu_ID = @St_ID AND S.Ex_ID = @Ex_ID
        AND S.Stu_Answers = Q.Correct_Answer;
    UPDATE Student_Exam
    SET
        Score = @Correct,
        Percentage = (CAST(@Correct AS FLOAT) / 10) * 100
    WHERE ST_ID = @St_ID AND Ex_ID = @Ex_ID;
    PRINT 'Exam submitted. Correct Answers: ' + CAST(@Correct AS VARCHAR(10));
END;

```

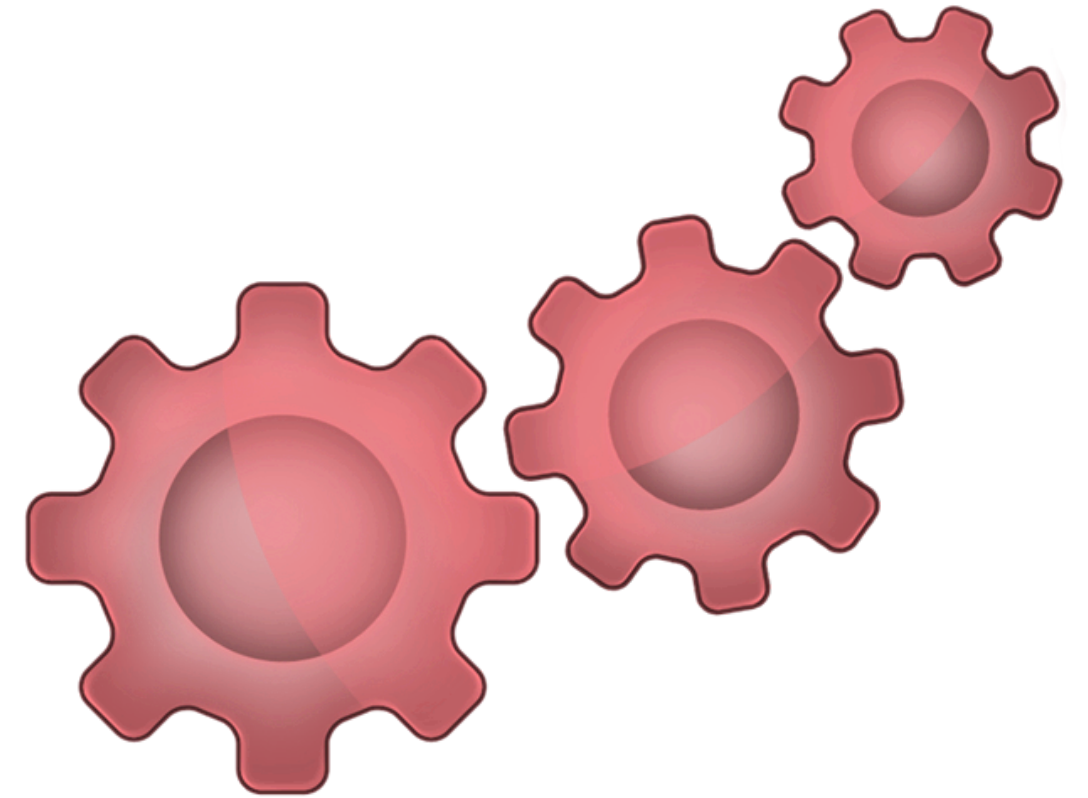

EXAM SP CHALLENGES



**Handling Random
Question Generation**



**Mapping Answers
to the Correct
Question Order**



**Preventing Multiple
Exam Submissions**

WEB APPLICATION

We developed a multi-page Streamlit web application to serve as a user-friendly interface for our SQL-based examination system. The application features a custom, professional UI and seamlessly navigates students through the entire exam process. It robustly connects to our database to execute stored procedures, automating everything from dynamic question generation to the instant correction .



WEB APPLICATION



Welcome To Our Examination System 🖱️

Click the button below to begin.

Start Now 🔗

© 2025 Examination System. All rights reserved.



Student Login 🔑

Student Email

raafat4@yahoo.com

Password

Login

[Back to Welcome Page](#)



Select Your Course

Choose a course to begin the exam:

Power BI

Start Exam



Take Your Exam, Raafat Elrais! 📝

Please choose the correct answers for the following questions:

Question 1: Which visual is best for showing trends over time?

- ☐ A) Bar Chart
- ☐ B) Pie Chart
- ☐ C) Line Chart
- ☐ D) Gauge

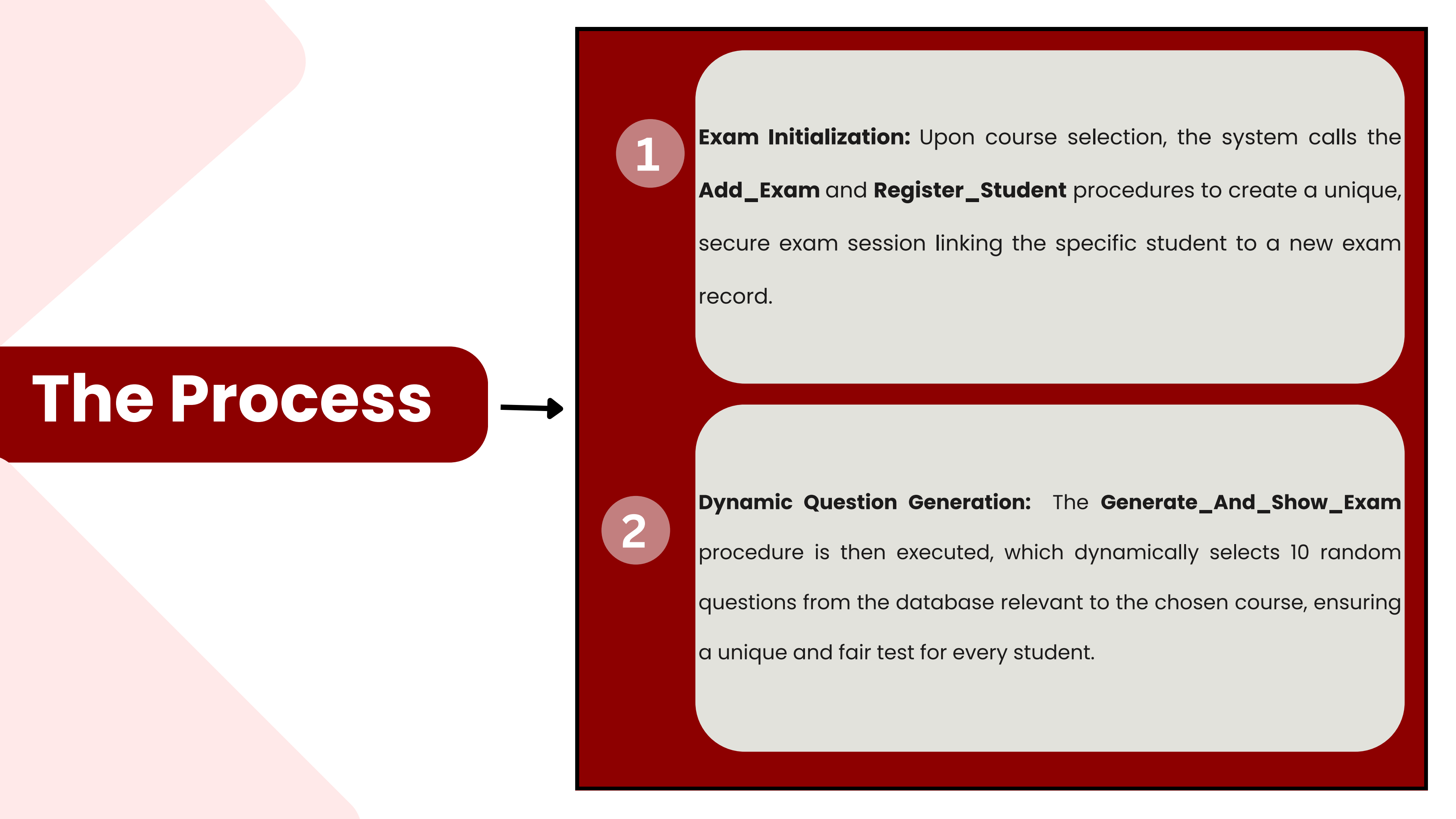
Your Score: 70 / 100

Grade: C (Good)

Thank You, Raafat Elrais! 😊

End and Return to Welcome Page

The Process



1

Exam Initialization: Upon course selection, the system calls the **Add_Exam** and **Register_Student** procedures to create a unique, secure exam session linking the specific student to a new exam record.

2

Dynamic Question Generation: The **Generate_And_Show_Exam** procedure is then executed, which dynamically selects 10 random questions from the database relevant to the chosen course, ensuring a unique and fair test for every student.

The Process

```
graph LR; A[The Process] --> B[3 Interactive Answering]; A --> C[4 Automated Scoring & Results];
```

3

Interactive Answering : The 10 questions and their choices are displayed in a clean web interface. After the student makes their selections, all answers are securely packaged and sent to the database upon submission.

4

Automated Scoring & Results: Finally, the **SubmitExamAnswers** procedure is called. It instantly corrects the submitted answers, calculates the final score, saves it to the Student_Exam table, and the result is immediately retrieved and displayed to the student.

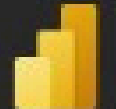
SQL SERVER REPORT SERVICES (SSRS)

Using SSRS in Visual Studio, we created interactive reports that transform raw database data into clear, actionable insights for students, instructors, and administrators.



SSRS

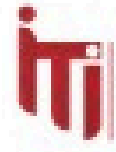
- Report that returns the students information according to Department No parameter. (GetStudentTopic)

 Power BI Report Server Home > Reports_ > Students Info

Topic Id

1 of 2 ?

100%

 Information Technology Institute

Students Info

Software Engineering

Name	Gender	City	Age	College	University	Grade Year
Mohamed Sayed	M	Suez	21	Faculty of Engineering	Tanta University	2023
Youssef Ibrahim	M	Mansoura	24	Faculty of Media & Mass Comm	Luxor University	2022
Amira Hussein	F	Mansoura	21	Faculty of Computer & Information Technology	Beni-Suef University	2024
Zainab Younis	F	Alexandria	23	Faculty of Business	Aswan University	2023
Mariam Sayed	F	Assiut	26	Faculty of Computer & Information Technology	Al-Azhar University	2020

SSRS

•Report that takes course ID and returns its topics.

Power BI Report Server Home > Reports_ > Courses By Topic

crs id

1 of 1

100%

Courses By Topic

Big Data

Topic Name
AI
Business development
Embedded Systems

(GetTopicbyCourses) Procedure

•Report that takes exam number and returns the Questions in it and choices [freeform report]

Power BI Report Server Home > Reports_ > Exam's Questions

Exam ID

1 of 1

100%

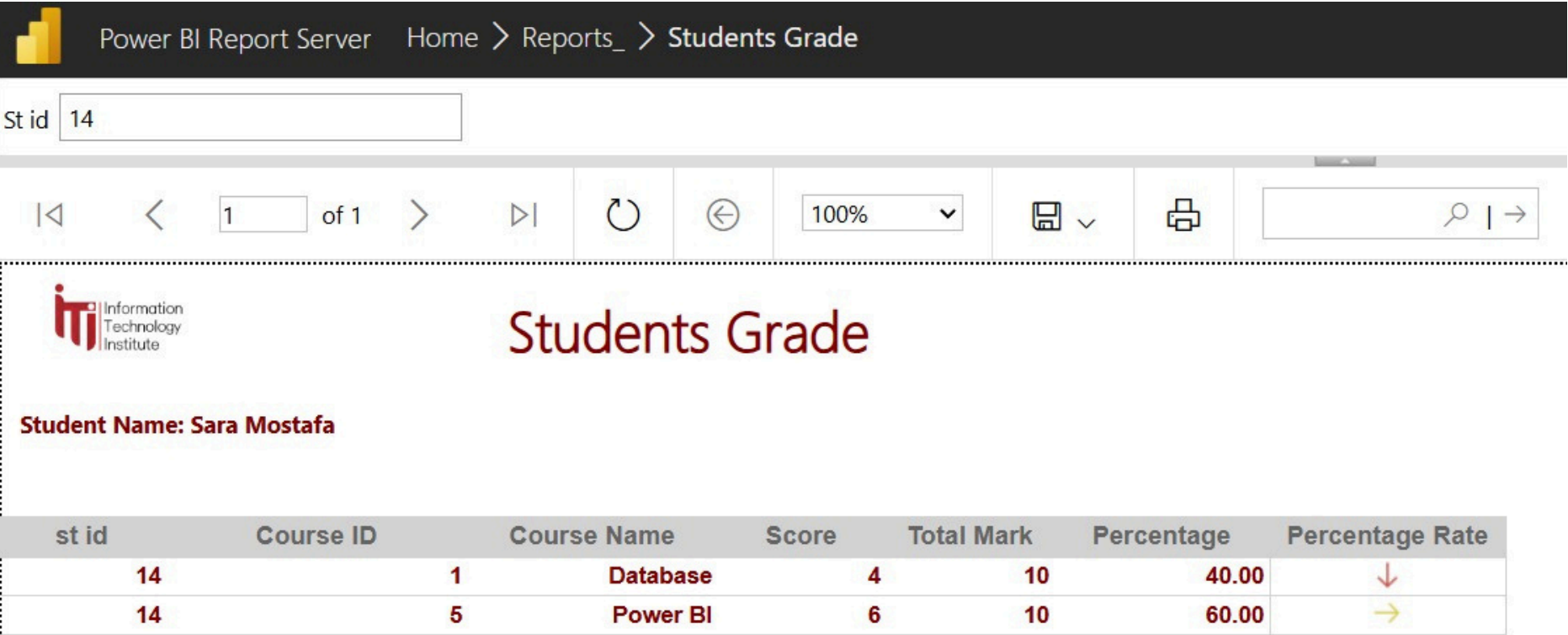
Qs ID	Qs Body	Option A	Option B	Option C	Option D
201	Which keyword is used to define a function in Python?	def	function	lambda	method
204	What symbol is used for comments in Python?	#	//	--	%
205	Which collection type allows duplicate elements?	list	set	tuple	dictionary
207	Which function displays output to the console?	print()	display()	echo()	show()
210	Which method removes the last element from a list?	remove()	pop()	delete()	clear()

(GetQuestions) Procedure

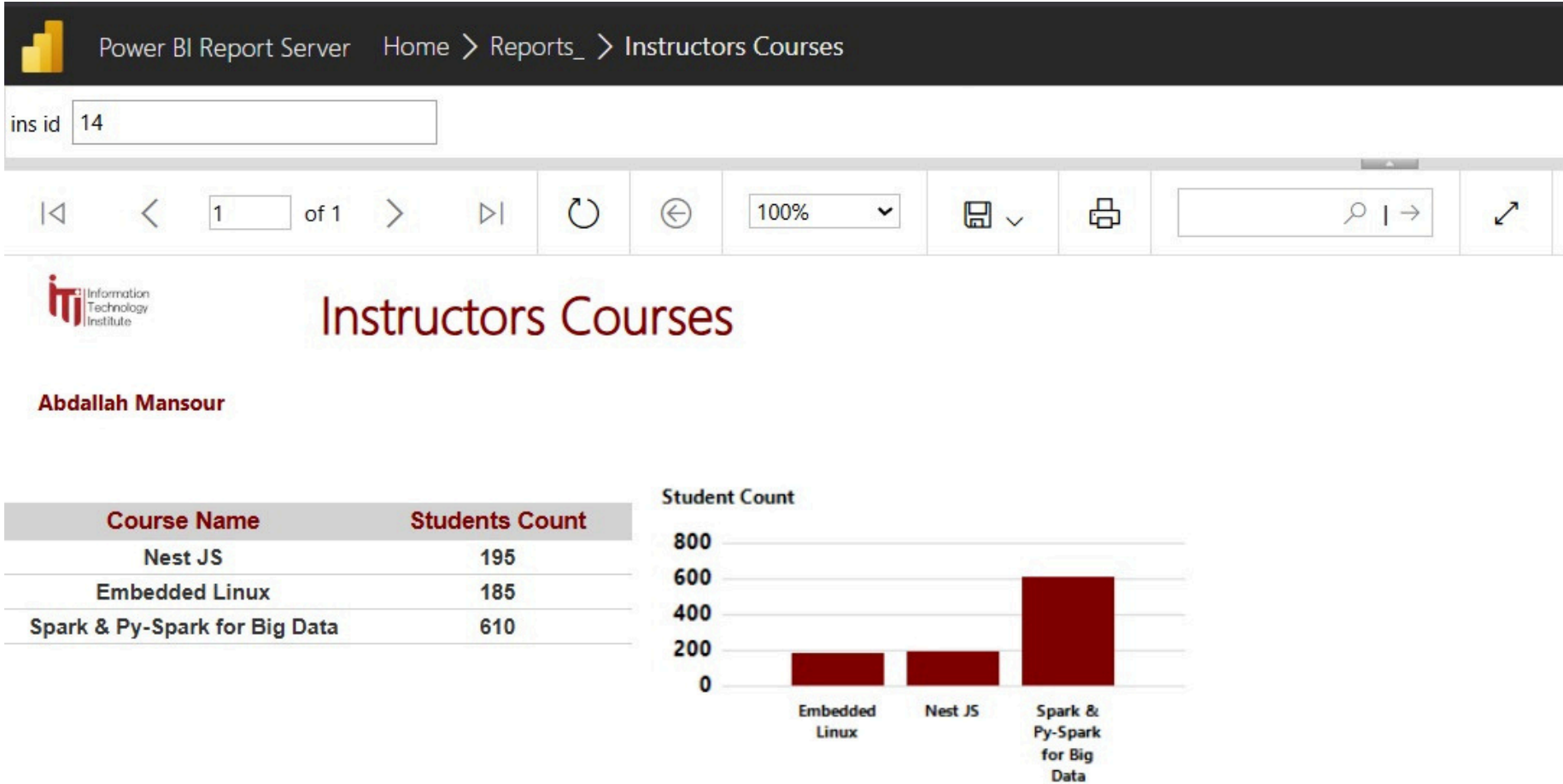
SSRS

• Report that takes the student ID and returns the grades of the student in all courses. %

• Report that takes the instructor ID and returns the name of the courses that he teaches and the number of students per course.



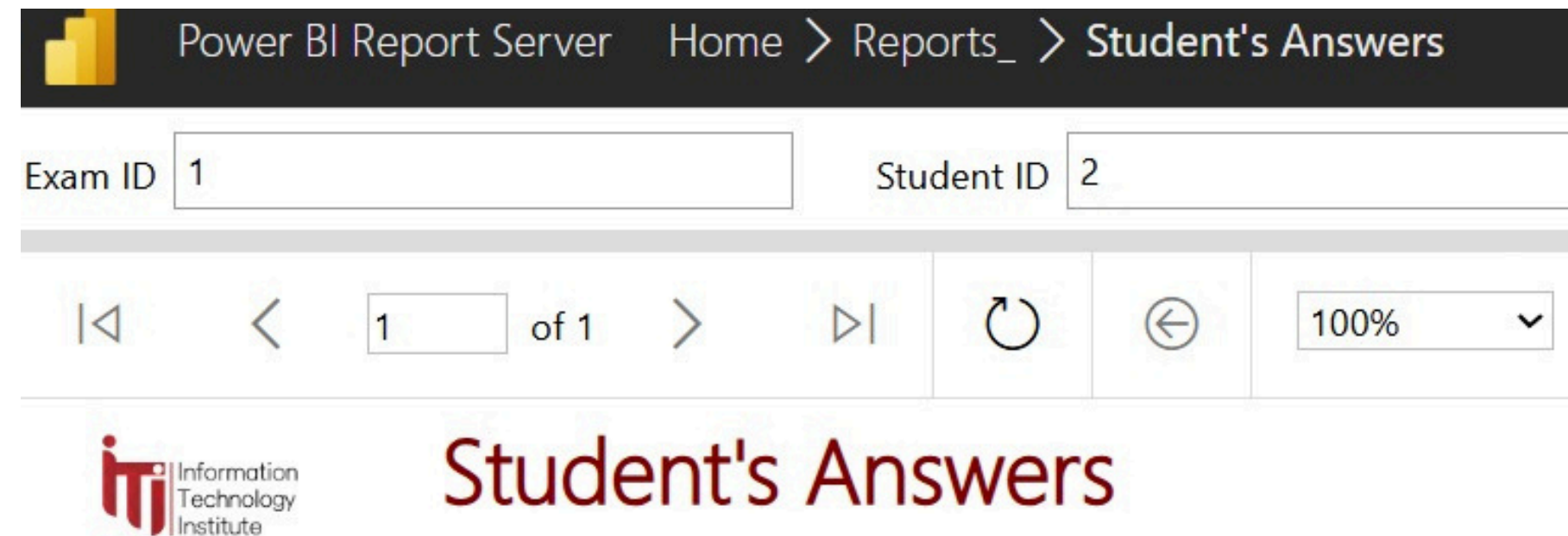
(GetStudentGrade) Procedure



(GetInstructorInfo) Procedure

SSRS

•Report that takes exam number and the student ID then returns the Questions in this exam with the student answers.



(GetStudentAnswers) Procedure

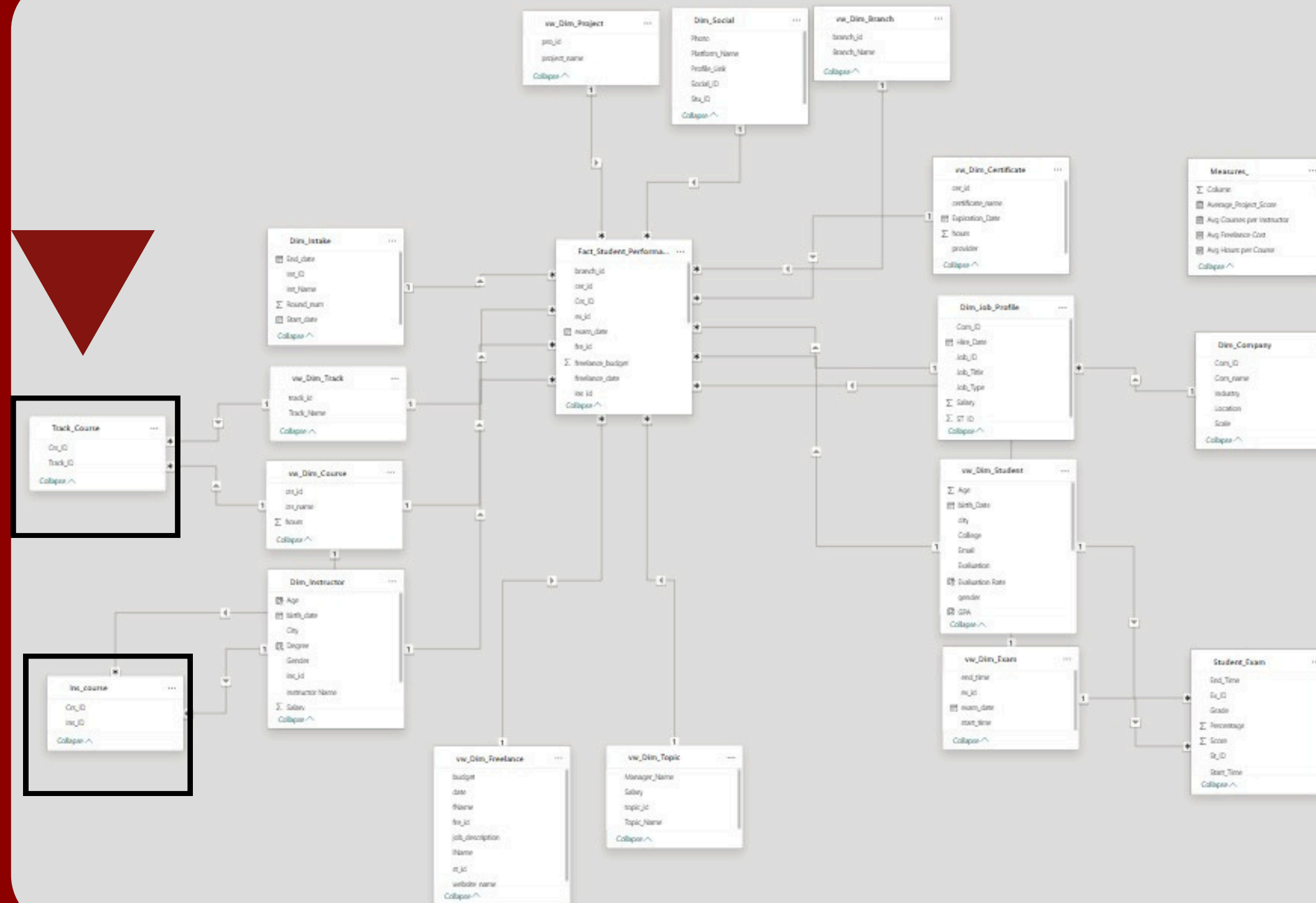
Exam ID: 1 | Student ID: 2

Qs Body	Student Answer
Which clause is used to filter records in SQL?	A
Which constraint ensures a column cannot have NULL values?	A
Which clause groups rows that have the same values?	A
What is the main key in a table called?	A
Which SQL function gives the current date?	A
Which command changes data in an existing	B

Data Model

- THE SYSTEM USES A **STAR SCHEMA** WITH A CENTRAL **FACT_STUDENT** A TABLE STORING KEY STUDENT METRICS. IT CONNECTS TO MULTIPLE DIMENSION TABLES, SUCH AS STUDENT, INSTRUCTOR, COURSE, EXAM, AND TRACK, AND PROVIDES CONTEXT FOR FILTERING AND ANALYSIS.
- A **BRIDGE TABLE** IS INCLUDED TO PROPERLY HANDLE **MANY-TO-MANY** RELATIONSHIPS AND ENSURE ACCURATE REPORTING. A DEDICATED MEASURES TABLE ORGANIZES DAX CALCULATIONS FOR BETTER PERFORMANCE AND MAINTENANCE.
- THIS DESIGN DELIVERS **FAST QUERIES, FLEXIBLE FILTERING,** AND A **SCALABLE FOUNDATION** FOR BUSINESS INTELLIGENCE.

Star Schema



POWER BI DASHBOARD

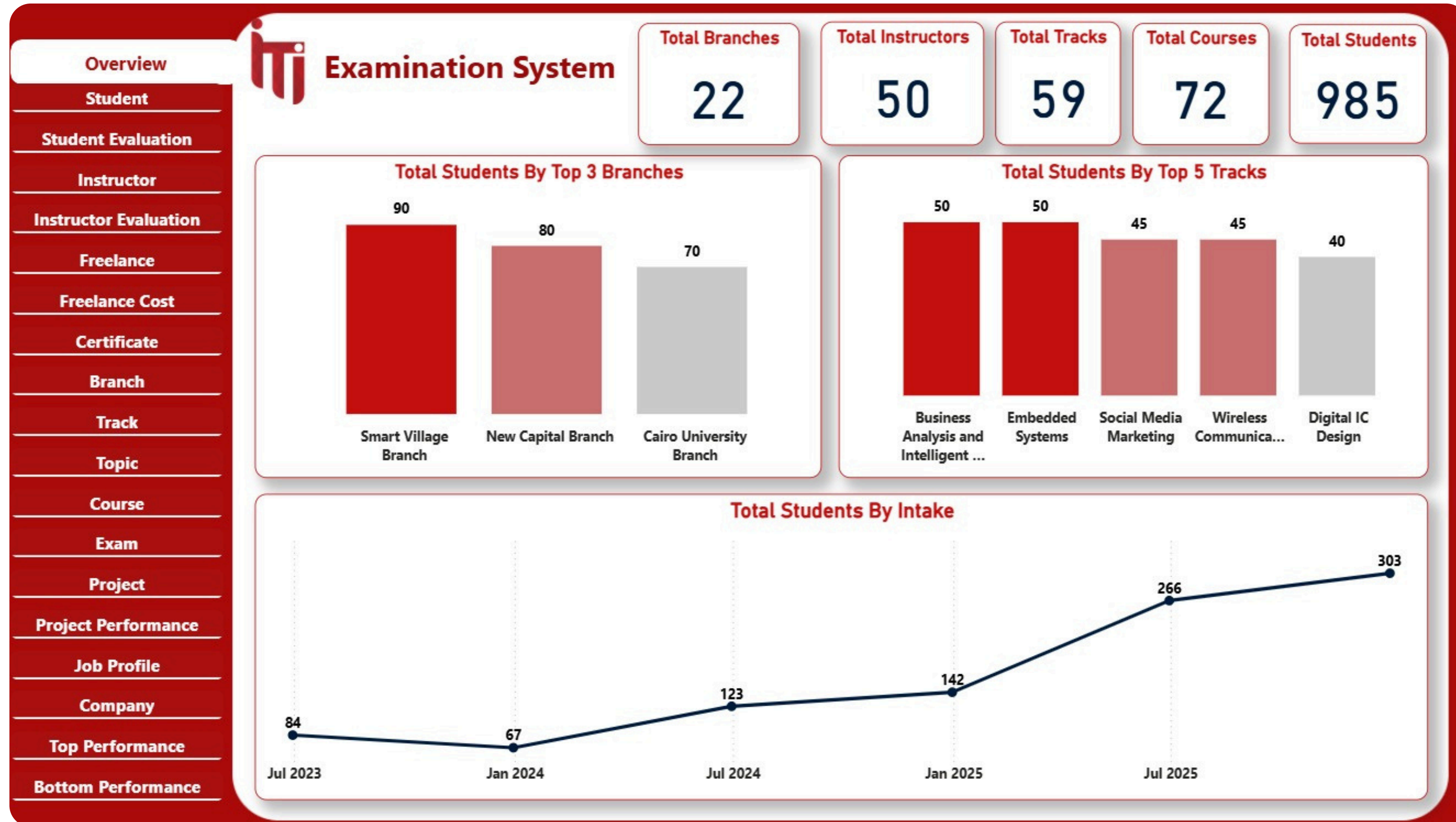


Power BI

POWER BI DASHBOARD

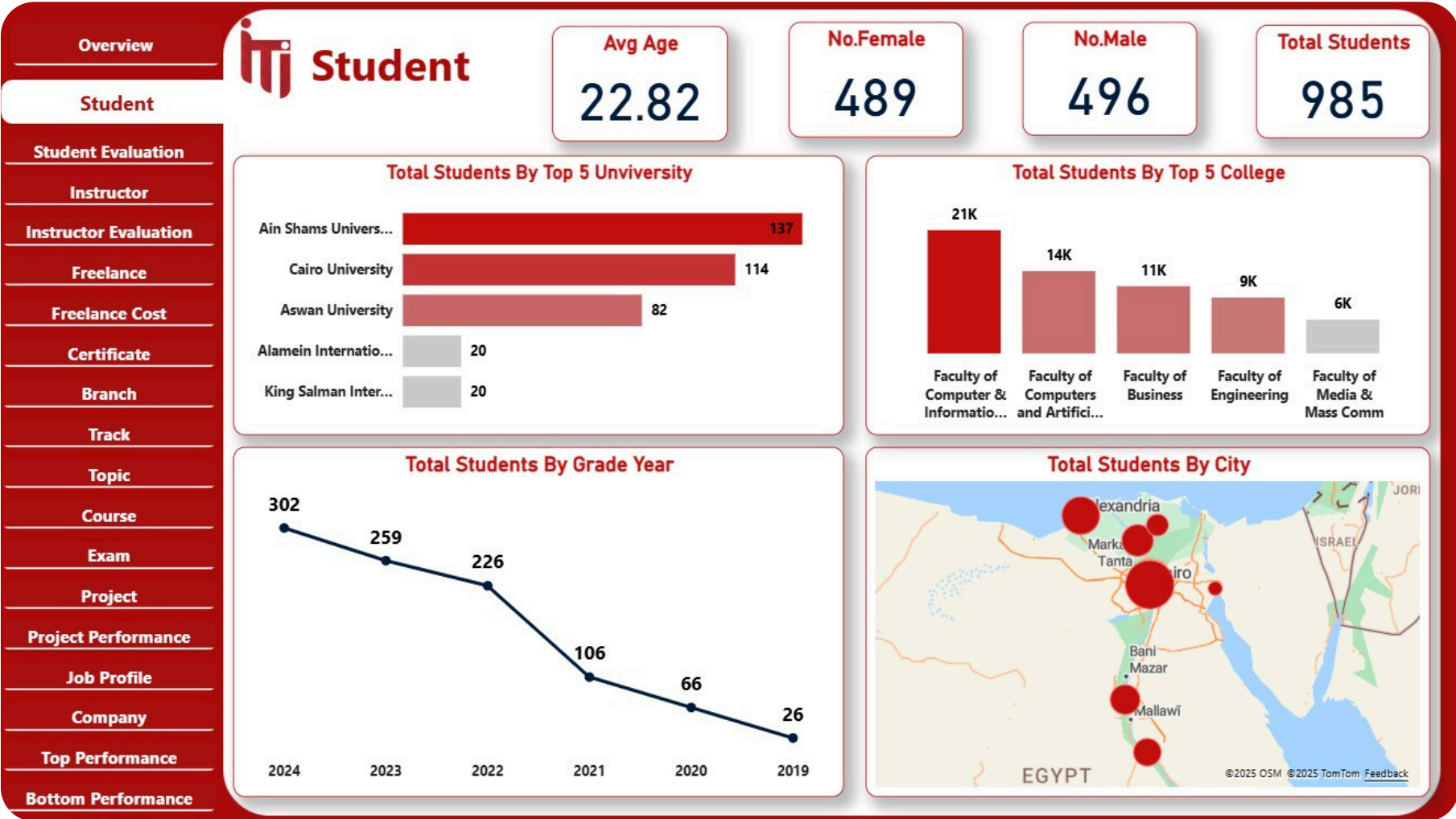
1

OVERVIEW

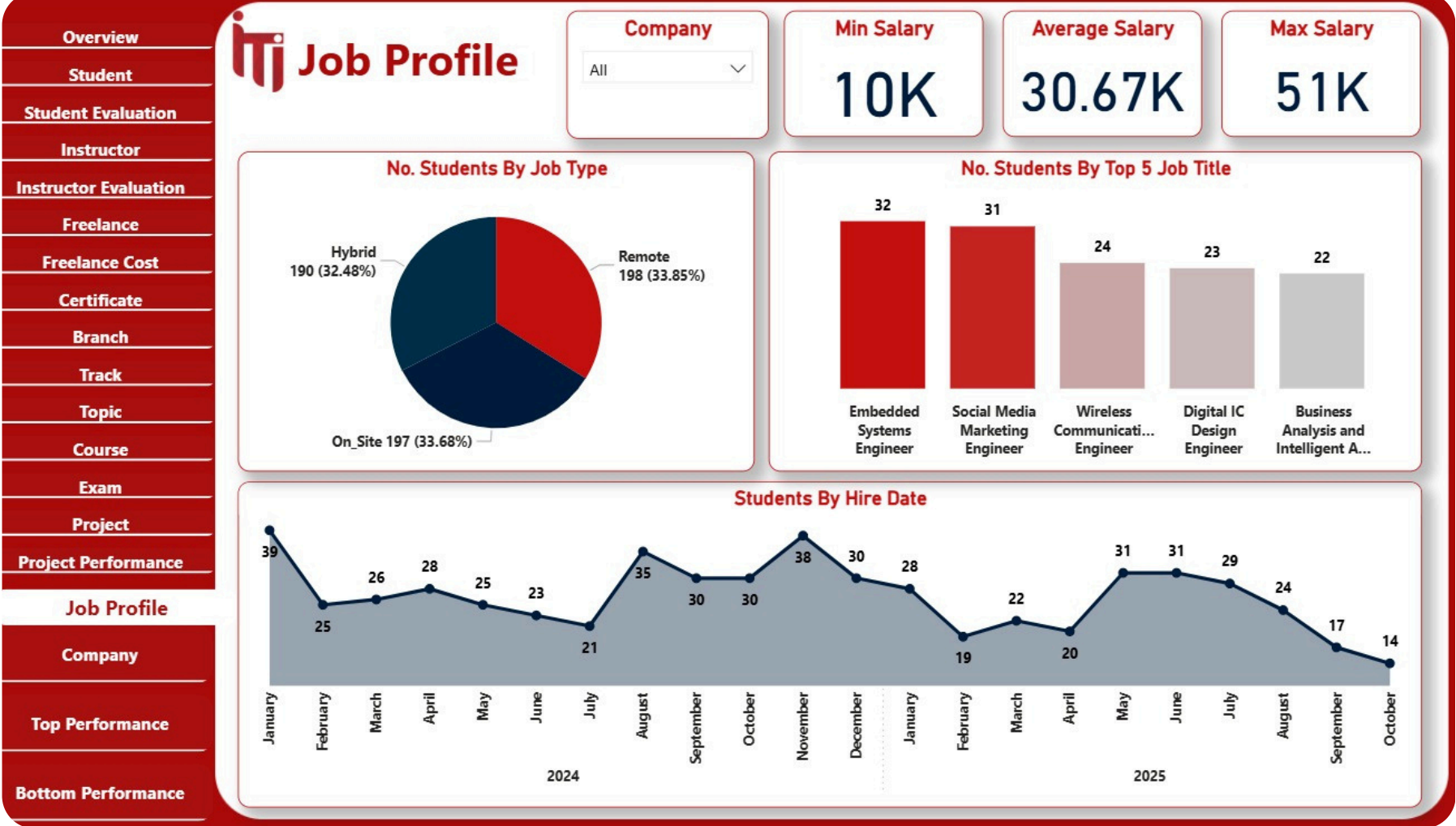


POWER BI DASHBOARD

2 STUDENT

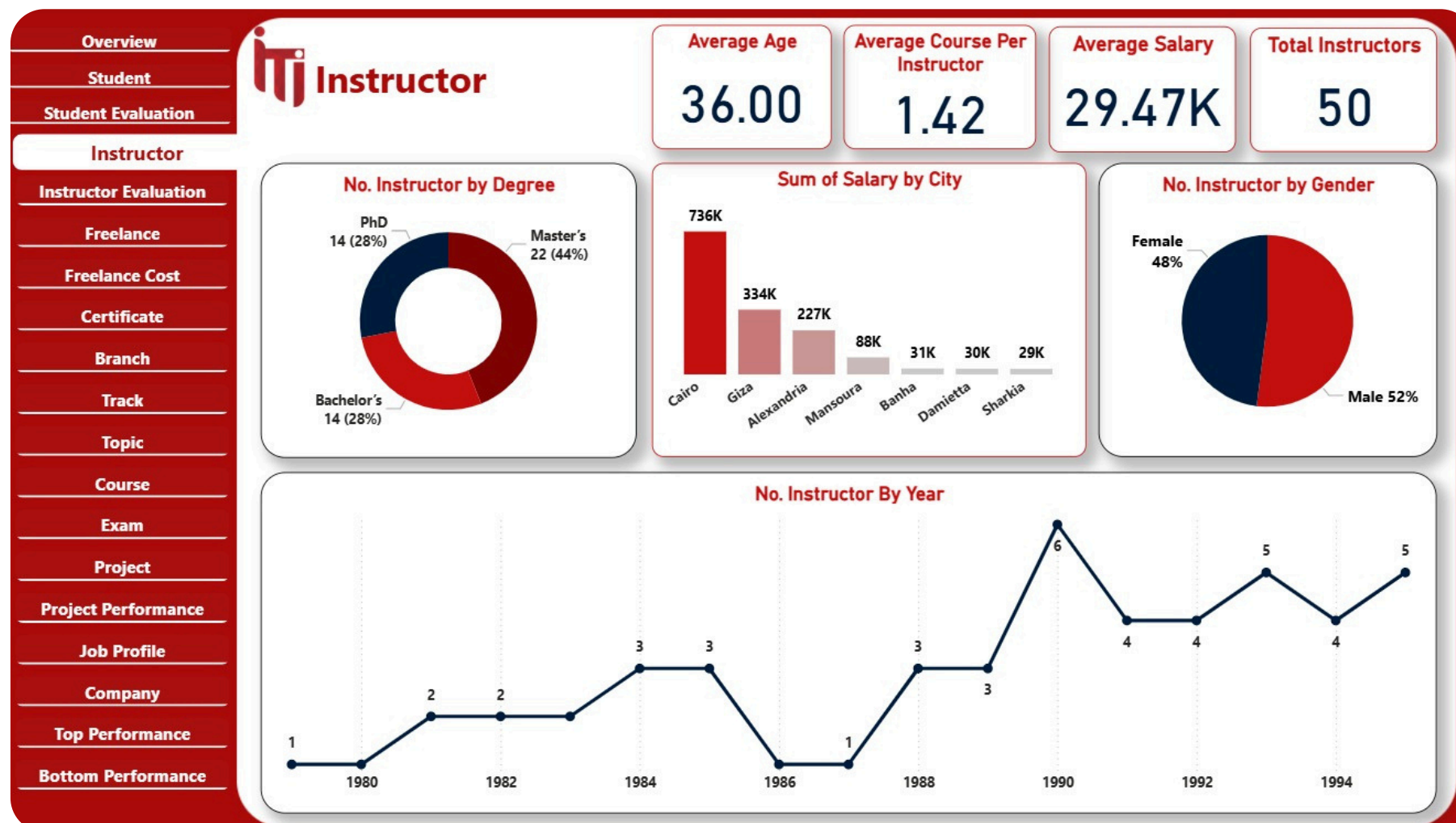


3 JOB PROFILE

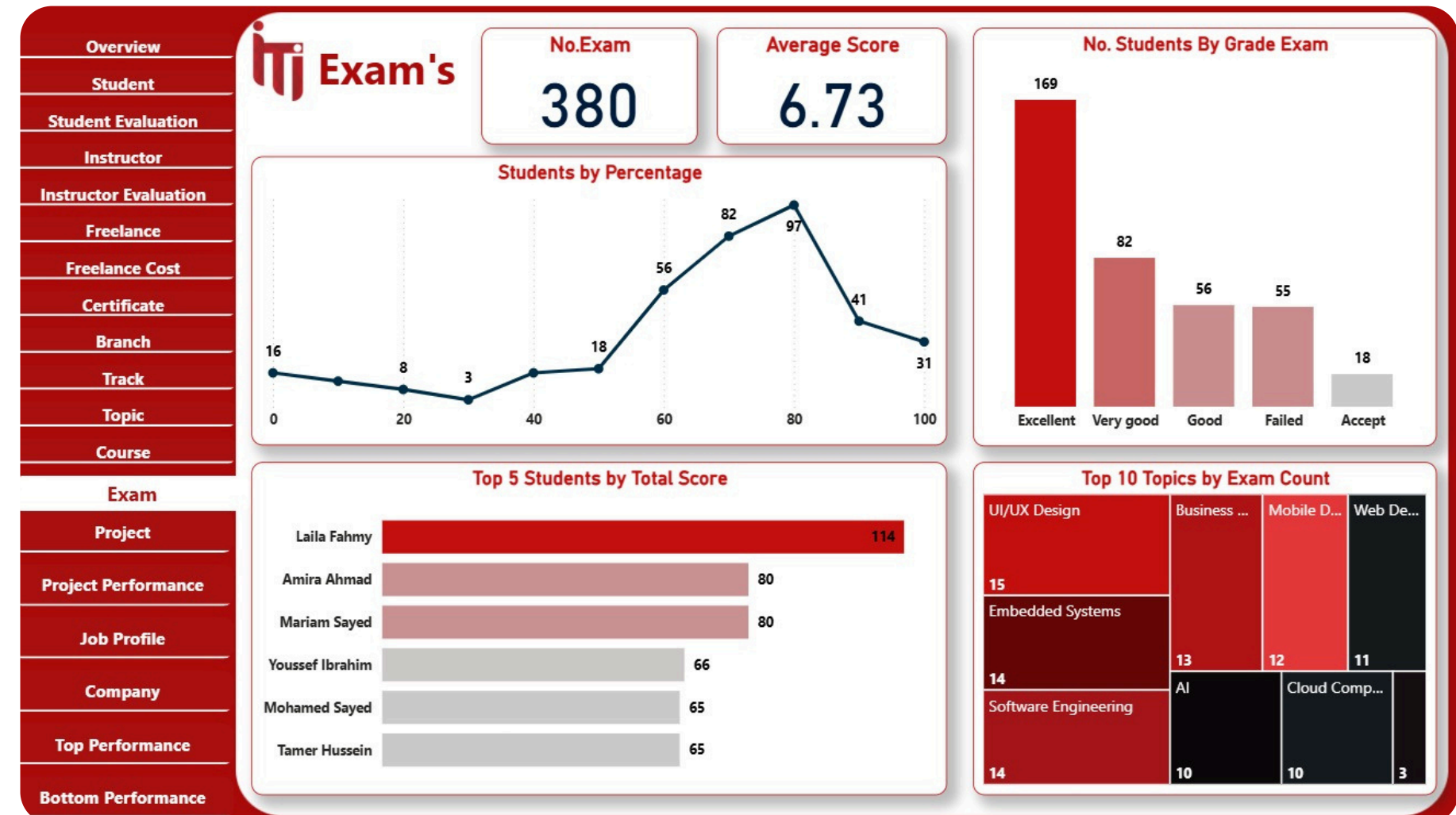


POWER BI DASHBOARD

4 INSTRUCTOR



5 EXAM



POWER APP

The Power Apps Examination System enables students to take exams digitally and submit answers through a simple interface. It connects to SQL Server for secure, real-time data handling. Power Apps was chosen for its fast development, mobile support, and Power BI integration.



Power Apps

The System Flow



1



Student Exam System

Ahmed Medhat

Login

2

Student ID: 3

Available Exams

1	1	>
2	2	>
3	3	>
4	4	>
5	5	>

3

Submit Exam

Which of the following creates a new table?

- ☒ CREATE TABLE
☐ ADD TABLE

Which SQL statement is used to rename a table?

- ☐ ALTER RENAME
☒ RENAME TABLE

4

Score: 3 / 10

30%

✖ Try Again!

true

CREATE TABLE

false

RENAME TABLE

The System Flow

Problems and Solutions

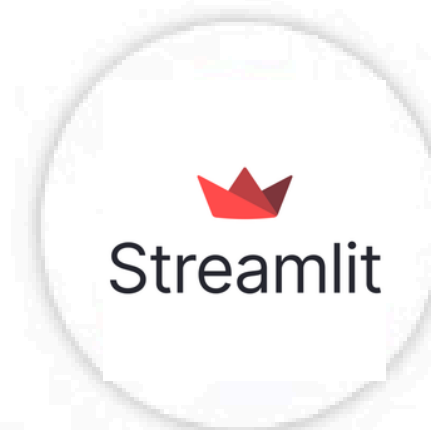
Problems

- **Unbalanced data impacting result accuracy.**
- **Foreign key deletion constraint issue.**
- **Difficulty choosing optimal data model**

Solutions

- **Applied asymmetric distribution for accuracy.**
- **Applied cascade and null strategies.**
- **Implemented efficient star schema design**

Used Tools



Future Plan

- **Instructor Question Configuration**
- **Smart Question Bank Expansion**
- **Adaptive Exam Generation**
- **Mobile Examination App**



QUESTIONS ??

The background features two large, light pink geometric shapes. One is a triangle on the left side, and the other is a quadrilateral on the right side, both pointing towards the center.

THANK YOU